

FINAL RESUBMISSION DOCUMENT

Project 1B: Custom API Integration

SAP S/4HANA ERP to FinSight Financial Analytics Platform

OpenAPI 3.0

102 Field Mappings

25 Test Scenarios

Runnable Prototype

Reconciliation Evidence

This report presents a clean, complete, and implementation-backed solution for designing an end-to-end API integration framework between SAP S/4HANA and a financial analytics platform.

Participant	Sachin / sachin-git01
GitHub Repository	https://github.com/sachin-git01/project2
Source System	SAP S/4HANA ERP
Target Platform	FinSight Financial Analytics Platform
Primary Output	Design documentation, API specs, mapping spec, monitoring plan, test plan, and runnable proof-of-concept
Validation Status	Prototype executed successfully; reconciliation status is RECONCILED

Table of Contents

1. Executive Summary

2. Problem Statement Alignment

3. Business and Technical Objectives

4. End-to-End Architecture

5. API Specification

6. Data Mapping and Transformation

7. Error Handling and Resilience

8. Reconciliation and Audit Controls

9. Monitoring and Operations

10. Runnable Prototype and Validation Evidence

11. Testing Strategy

12. Submission Deliverables

13. Appendix A: Full Mapping Table

14. Appendix B: Full Test Scenario Table

1. Executive Summary

Meridian Manufacturing uses SAP S/4HANA as its ERP system of record but needs faster and more reliable financial analytics in FinSight. This solution removes manual exports by creating a resilient integration layer that extracts SAP data, transforms it into canonical financial schemas, loads it to FinSight, and verifies every batch through reconciliation.

The design follows enterprise integration best practices: SAP-supported extraction methods, Kafka-based decoupling, idempotent destination APIs, DLQ isolation, circuit breakers, audit lineage, and operational dashboards.

4 hr

Standard freshness SLA

Finance analytics available within 4 hours for standard domains.

102

Field mappings

Field-level source, target, transformation, validation, and error handling rules.

25

Test scenarios

Functional, non-functional, failure, security, and reconciliation tests.

0

Duplicate tolerance

Idempotency keys prevent duplicate FinSight writes.

2. Problem Statement Alignment

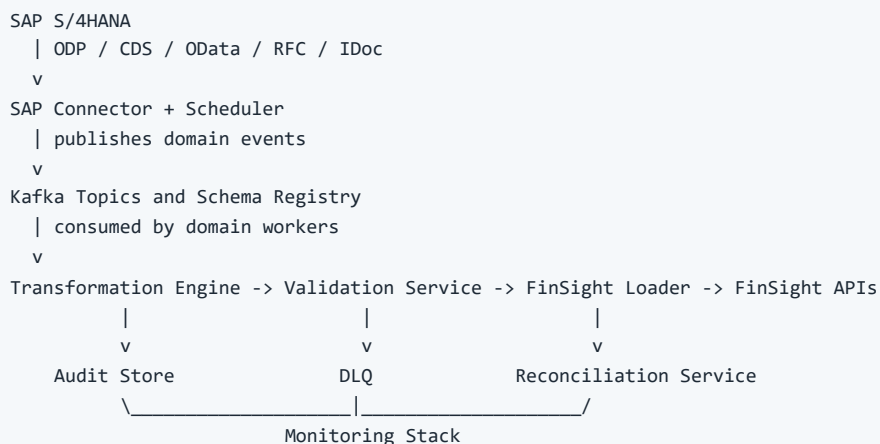
Project Requirement	Solution Coverage	Evidence
Custom API integration framework	Source API facade, Kafka integration layer, transformation engine, validation service, FinSight loader, reconciliation, and monitoring.	Architecture + APIs + prototype
SAP S/4HANA source system	Uses ACDOCA, BKPF, ODP, CDS, OData, RFC, IDoc, and SAP master data concepts.	Source endpoint table and mappings
Financial analytics platform	Defines FinSight batch ingestion endpoints and canonical financial payloads.	Destination API table
API specification	OpenAPI 3.0 YAML files for SAP and FinSight.	api/API_SAP_Source.yaml and api/API_FinSight_Destination.yaml

Project Requirement	Solution Coverage	Evidence
Data extraction	ODP/CDS delta strategy plus runnable CSV extraction simulation.	Prototype and extraction design
Transformation logic	102 documented mappings plus executable Python transformation logic.	Appendix A and transform.py
Communication protocols	HTTPS/OData, RFC/SNC, Kafka, OAuth 2.0, TLS 1.2+, and API gateway controls.	Architecture and security sections

3. Business and Technical Objectives

Objective	Implementation	Success Measure
Improve finance visibility	Automated SAP-to-FinSight data flow	Dashboards refreshed within SLA
Reduce manual reconciliation	Batch reconciliation service	Counts, totals, and checksums match
Protect SAP stability	Delta extraction, package limits, off-peak full loads	No production performance degradation
Ensure auditability	Lineage fields from SAP document to FinSight record	Every target record traceable
Handle failures safely	Retries, circuit breakers, DLQ, business exception queues	No data loss during transient failures

4. End-to-End Architecture



Architecture Components

Component	Responsibility
Scheduler	Triggers full loads, deltas, retries, and reconciliation jobs.
SAP Connector	Reads SAP source data through supported enterprise interfaces.
Kafka	Buffers messages, supports replay, and isolates SAP from FinSight outages.
Transformation Engine	Converts SAP records to canonical FinSight schemas.
Validation Service	Checks mandatory fields, dates, currency, and master data references.
FinSight Loader	Loads records using batch upsert APIs and idempotency keys.
Reconciliation Service	Compares source and target counts, amounts, checksums, and lineage.
Monitoring Stack	Tracks freshness, throughput, errors, DLQ, lag, and circuit breakers.

5. API Specification

SAP Source APIs

ID	Domain	Endpoint	Extraction	SAP Source
SRC-001	General Ledger	/sap/v1/general-ledger/journal-entries	ODP delta	ACDOCA/BKPF
SRC-002	Accounts Payable	/sap/v1/accounts-payable/open-items	ODP delta	ACDOCA/LFA1
SRC-003	Accounts Receivable	/sap/v1/accounts-receivable/open-items	ODP delta	ACDOCA/KNA1
SRC-004	Cost Centre	/sap/v1/master-data/cost-centres	CDS/OData	CSKS/CSKT
SRC-005	Profit Centre	/sap/v1/master-data/profit-centres	CDS/OData	CEPC/CEPCT
SRC-006	Material Ledger	/sap/v1/material-ledger/movements	CDS delta	MLDOC/CKMLCR
SRC-007	Purchase Orders	/sap/v1/procurement/purchase-orders	OData/CDS	EKKO/EKPO/EKBE
SRC-008	Sales Orders	/sap/v1/sales/sales-orders	OData/CDS	VBAP/VBAP/VBFA
SRC-009	Fixed Assets	/sap/v1/fixed-assets/assets	CDS	ANLA/ANLC
SRC-010	Bank Statements	/sap/v1/cash/bank-statements	IDoc/CDS	FEBKO/FEBEP
SRC-011	Budget vs Actual	/sap/v1/controlling/budget-actuals	CDS	ACDOCA/ACDOCP
SRC-012	Inventory	/sap/v1/inventory/stock-movements	ODP/CDS	MATDOC/MARA

FinSight Destination APIs

ID	Domain	Endpoint
DST-001	General Ledger	/finsight/v1/journal-entries:batchUpsert
DST-002	Accounts Payable	/finsight/v1/payables:batchUpsert
DST-003	Accounts Receivable	/finsight/v1/receivables:batchUpsert
DST-004	Cost Centre	/finsight/v1/cost-centres:batchUpsert
DST-005	Profit Centre	/finsight/v1/profit-centres:batchUpsert
DST-006	Material Ledger	/finsight/v1/material-movements:batchUpsert
DST-007	Purchase Orders	/finsight/v1/purchase-orders:batchUpsert
DST-008	Sales Orders	/finsight/v1/sales-orders:batchUpsert
DST-009	Fixed Assets	/finsight/v1/fixed-assets:batchUpsert
DST-010	Bank Statements	/finsight/v1/bank-statements:batchUpsert
DST-011	Budget vs Actual	/finsight/v1/budget-actuals:batchUpsert
DST-012	Inventory	/finsight/v1/inventory:batchUpsert

API design includes OAuth 2.0 authentication, cursor pagination, delta tokens, rate-limit headers, Retry-After handling, common error envelope, and mandatory Idempotency-Key for writes.

6. Data Mapping and Transformation

The project includes 102 field-level mappings across 12 domains. Each mapping includes source field, target field, transformation rule, validation, and error handling.

Domain	Mapping Count	Coverage
AP	102	100%
10		
Source field, target field, transformation, validation, and error handling documented		
AR		
10		
Source field, target field, transformation, validation, and error handling documented		
Bank Statement		
7		

Domain	Mapping Count	Coverage
Source field, target field, transformation, validation, and error handling documented		
BudgetActual		
8		
Source field, target field, transformation, validation, and error handling documented		
Cost Centre		
6		
Source field, target field, transformation, validation, and error handling documented		
Fixed Assets		
6		
Source field, target field, transformation, validation, and error handling documented		
GL		
15		
Source field, target field, transformation, validation, and error handling documented		
Inventory		
10		
Source field, target field, transformation, validation, and error handling documented		
Material Ledger		
8		
Source field, target field, transformation, validation, and error handling documented		
Profit Centre		
6		
Source field, target field, transformation, validation, and error handling documented		
Purchase Order		
8		
Source field, target field, transformation, validation, and error handling documented		
Sales Order		
8		
Source field, target field, transformation, validation, and error handling documented		

Core Transformation Rules

Rule	Decision
Date conversion	SAP YYYYMMDD values are converted to ISO YYYY-MM-DD.
Currency conversion	Transaction currency is validated against ISO 4217 and converted to INR using point-in-time rates.
Fiscal period handling	Periods 013-016 are mapped as special periods while preserving audit flags.
Idempotency	Keys are generated from tenant, domain, company code, fiscal year, document number, and line item.
Lineage	Every target record carries source system, source table, source document, and batch ID.

7. Error Handling and Resilience

Code	Scenario	Default Action
ERR-EXT-001	SAP RFC timeout	Retry with exponential backoff
ERR-EXT-003	ODP subscription invalidated	Halt domain and alert integration team
ERR-MAP-001	Mandatory field missing	Default if allowed else DLQ
ERR-MAP-002	Invalid date format	Correct if possible else DLQ
ERR-MAP-003	Missing master data reference	Business exception queue

Code	Scenario	Default Action
ERR-MAP-004	Invalid currency	Typo correction else DLQ
ERR-LOAD-001	FinSight HTTP 429	Honor Retry-After and throttle
ERR-LOAD-004	Duplicate entry	Compare payload hash and skip identical
ERR-RECON-001	Record count mismatch	Generate break report
ERR-SYS-001	Kafka broker unreachable	Circuit breaker and P1 alert

Resilience Patterns

Pattern	Usage
Exponential backoff with jitter	Retries transient SAP and FinSight failures without overwhelming recovering services.
Circuit breaker	Stops repeated calls to unhealthy SAP, FinSight, or Kafka dependencies.
Dead letter queue	Stores records that fail validation or mapping after controlled handling.
Idempotent receiver	Prevents duplicate target writes during retry or replay.
Bulkhead isolation	Prevents one failed domain from blocking other domains.

8. Reconciliation and Audit Controls

Control	Purpose	Expected Result
Record count	Source count equals accepted plus rejected records.	No unexplained missing records.
Debit-credit total	Validates financial balance after transformation.	Debit total equals credit total for balanced batches.
Checksum	Detects accidental changes in business keys or amounts.	Stable hash per batch.
Lineage	Connects FinSight record back to SAP source document.	Audit-ready traceability.

Generated Reconciliation Evidence

```
{
  "batchId": "BATCH-GL-20260629-0930",
  "domain": "GENERAL_LEDGER",
  "sourceRecordCount": 8,
  "acceptedRecordCount": 6,
  "rejectedRecordCount": 2,
  "debitTotalInr": 253125.0,
  "creditTotalInr": 253125.0,
  "netAmountInr": 0.0,
  "checksum": "8676c1adfc4a3254d74b7435e4948929717a01668e919b648920480fca4665ef",
  "status": "RECONCILED"
}
```

9. Monitoring and Operations

Panel	Title	Metric Focus	Alert
MON-001	Pipeline Health	Domain freshness, failures, last successful load	P1 when RED
MON-002	Throughput	records/sec by domain	Warn below 50% baseline
MON-003	Latency	P95/P99 API latency	Warn above 5 seconds
MON-004	Error Rate	errors / requests	Critical above 5%
MON-005	DLQ Depth	failed record backlog	P1 above 500
MON-006	Reconciliation	PASS/WARN/BREAK per batch	P2 on break
MON-007	SAP Health	RFC pool, response time, work processes	Critical above 90% pool
MON-008	FinSight API	rate limit, response time, token expiry	Warn below 10% quota
MON-009	Data Freshness	now - last load	Alert on SLA breach
MON-010	Resources	CPU, memory, disk, network	P2 above 90%

Panel	Title	Metric Focus	Alert
MON-011	Kafka Lag	consumer group lag	P2 above 50K
MON-012	Circuit Breakers	open/half-open/closed	P2 if open

10. Runnable Prototype and Validation Evidence

A runnable Python prototype is included to prove the integration behavior. It reads SAP-style GL data, validates records, transforms valid rows into FinSight JSON, routes invalid records to DLQ, and generates a reconciliation report.

Run Commands

```
$env:PYTHONPATH='src'
python -m sap_finsight_integration.cli
python -m unittest discover -s tests -p '*unittest.py'
```

Observed Result

Item	Result
Batch ID	BATCH-GL-20260629-0930
Status	RECONCILED
Accepted records	6
DLQ records	2
Automated tests	2 tests passed using Python unittest

Sample SAP Input

```
company_code,fiscal_year,fiscal_period,document_number,line_item,gl_account,cost_centre,profit_centre,posting_date,currency,amount,debit_credit
1000,2026,003,5100001001,1,400000,CC100,PC100,20260628,INR,125000.00,S
1000,2026,003,5100001001,2,110000,CC100,PC100,20260628,INR,125000.00,H
1000,2026,003,5100001002,1,500000,CC200,PC200,20260628,USD,1000.00,S
1000,2026,003,5100001002,2,210000,CC200,PC200,20260628,USD,1000.00,H
```

Load Result

```
{
  "batchId": "BATCH-GL-20260629-0930",
  "status": "COMPLETED",
  "acceptedCount": 6,
  "rejectedCount": 0
}
```

11. Testing Strategy

The test plan covers functional correctness, performance, concurrency, failure recovery, security, and reconciliation controls. The complete test table is included in Appendix B.

Category	Coverage
Functional	GL, AP, AR, master data, hierarchy, P2P, bank statements, budget, and full reconciliation.
Non-functional	Peak load, concurrency, latency, scalability, and endurance.
Failure	SAP RFC failure, FinSight 429, Kafka failure, malformed records, and network partition.
Security	OAuth expiry, invalid token, TLS, and encryption.
Reconciliation	Deliberate amount variance and missing master data.

12. Submission Deliverables

Deliverable	Artifact	Status
Project overview	README.md	Complete
Requirements	docs/D0_Requirements_v1.md	Complete
Architecture	docs/D1_Integration_Architecture_v1.md and diagrams/*.mmd	Complete
API specs	api/API_SAP_Source.yaml and api/API_FinSight_Destination.yaml	Complete
Mappings	mappings/MAP_Finance_AllDomains_v1.md	Complete, 102 mappings
Error handling	docs/D4_Error_Handling_Framework_v1.md	Complete
Reconciliation	docs/D5_Reconciliation_Specification_v1.md	Complete
Monitoring	monitoring/D6_Monitoring_Dashboard_v1.md	Complete
Test plan	tests/D7_Integration_Test_Plan_v1.md	Complete, 25 scenarios
Runnable prototype	src/, data/sample/, outputs/, tests/	Complete and verified

Appendix A: Full Mapping Table

Full 102-field mapping evidence. Font is compact only for appendix density; main explanation appears in Section 6.

ID	Domain	Source Field	Target Field	Transformation	Validation	Error Handling
MAP-GL-001	GL	ACDOCA-RBUKRS	companyCode	Trim and preserve 4-char code	Required, known company	ERR-MAP-001
MAP-GL-002	GL	ACDOCA-GJAHR	fiscalYear	String YYYY	Required, 4 digits	ERR-MAP-001
MAP-GL-003	GL	ACDOCA-BELNR	sourceDocument	Preserve SAP document number	Required	ERR-MAP-001
MAP-GL-004	GL	ACDOCA-DOCLN	lineItem	Left-pad to 3 digits	Required	ERR-MAP-001
MAP-GL-005	GL	ACDOCA-RACCT	glAccount	Strip leading spaces	Required, chart lookup	ERR-MAP-003
MAP-GL-006	GL	ACDOCA-RCNTR	costCentre	Null if blank	Must exist when populated	ERR-MAP-003
MAP-GL-007	GL	ACDOCA-PRCTR	profitCentre	Null if blank	Must exist when populated	ERR-MAP-003
MAP-GL-008	GL	BKPF-BUDAT	postingDate	Convert YYYYMMDD to ISO date	Valid date	ERR-MAP-002
MAP-GL-009	GL	BKPF-BLDAT	documentDate	Convert YYYYMMDD to ISO date	Valid date	ERR-MAP-002
MAP-GL-010	GL	ACDOCA-HSL	amountInFunctionalCurrency	Decimal(18,2)	Numeric	ERR-MAP-006
MAP-GL-011	GL	ACDOCA-WSL	amountInTransactionCurrency	Decimal(18,2)	Numeric	ERR-MAP-006
MAP-GL-012	GL	ACDOCA-RWCUR	transactionCurrency	ISO currency	ISO 4217	ERR-MAP-004
MAP-GL-013	GL	ACDOCA-DRCRK	debitCredit	S=DEBIT, H=CREDIT	Enum	ERR-MAP-001
MAP-GL-014	GL	ACDOCA-POPER	fiscalPeriod	Map 001-012, 013-016 as special	001-016	ERR-MAP-001
MAP-GL-015	GL	Composite	sourceBusinessKey	company;document;year;item	Unique	ERR-LOAD-004
MAP-AP-001	AP	ACDOCA-LIFNR	vendorId	Strip leading zeroes for display, retain raw	Required	ERR-MAP-001
MAP-AP-002	AP	LFA1-NAME1	vendorName	Trim	Required for enrichment	ERR-MAP-003
MAP-AP-003	AP	LFA1-STCD3	vendorGstin	Uppercase	GSTIN pattern when India vendor	ERR-MAP-004
MAP-AP-004	AP	ACDOCA-BELNR	invoiceDocument	Preserve	Required	ERR-MAP-001
MAP-AP-005	AP	ACDOCA-AUGBL	clearingDocument	Null if open	Optional	DLQ only if invalid format
MAP-AP-006	AP	ACDOCA-ZFBDT	baselineDate	YYYYMMDD to ISO	Valid date	ERR-MAP-002
MAP-AP-007	AP	ACDOCA-ZBD1T	paymentTermsDays	Integer	>=0	ERR-MAP-006
MAP-AP-008	AP	Derived	ageingBucket	days past due: 0-30/31-60/61-90/90+	Required	ERR-MAP-006
MAP-AP-009	AP	ACDOCA-TSL	openAmount	Decimal	Numeric	ERR-MAP-006
MAP-AP-010	AP	ACDOCA-RWCUR	currency	ISO currency	ISO 4217	ERR-MAP-004
MAP-AR-001	AR	ACDOCA-KUNNR	customerId	Strip leading zeroes for display	Required	ERR-MAP-001
MAP-AR-002	AR	KNA1-NAME1	customerName	Trim	Required for enrichment	ERR-MAP-003
MAP-AR-003	AR	KNA1-STCD3	customerGstin	Uppercase	GSTIN pattern when India customer	ERR-MAP-004
MAP-AR-004	AR	ACDOCA-BELNR	invoiceDocument	Preserve	Required	ERR-MAP-001
MAP-AR-005	AR	ACDOCA-ZFBDT	baselineDate	YYYYMMDD to ISO	Valid date	ERR-MAP-002
MAP-AR-006	AR	KNKK-KLIMK	creditLimit	Decimal	>=0	ERR-MAP-006
MAP-AR-007	AR	KNKK-SAUFT	creditExposure	Decimal	>=0	ERR-MAP-006
MAP-AR-008	AR	ACDOCA-MANST	dunningLevel	Map SAP level to LOW/MEDIUM/HIGH	Known code	ERR-MAP-001
MAP-AR-009	AR	Derived	ageingBucket	days past due	Required	ERR-MAP-006
MAP-AR-010	AR	ACDOCA-TSL	openAmount	Decimal	Numeric	ERR-MAP-006
MAP-CC-001	Cost Centre	CSKS-KOKRS	controllingArea	Preserve	Required	ERR-MAP-001
MAP-CC-002	Cost Centre	CSKS-KOSTL	costCentre	Preserve	Required, unique	ERR-LOAD-004
MAP-CC-003	Cost Centre	CSKT-KTEXT	costCentreName	Trim	Required	ERR-MAP-001
MAP-CC-004	Cost Centre	CSKS-DATAB	validFrom	YYYYMMDD to ISO	Valid date	ERR-MAP-002
MAP-CC-005	Cost Centre	CSKS-DATBI	validTo	YYYYMMDD to ISO	>= validFrom	ERR-MAP-002
MAP-CC-006	Cost Centre	SETNODE	hierarchyPath	Flatten to Level1-Level7	All parents exist	ERR-MAP-003
MAP-PC-001	Profit Centre	CEPC-KOKRS	controllingArea	Preserve	Required	ERR-MAP-001
MAP-PC-002	Profit Centre	CEPC-PRCTR	profitCentre	Preserve	Required, unique	ERR-LOAD-004
MAP-PC-003	Profit Centre	CEPC-KTEXT	profitCentreName	Trim	Required	ERR-MAP-001
MAP-PC-004	Profit Centre	CEPC-SEGMENT	segment	Map to reporting segment	Known segment	ERR-MAP-003
MAP-PC-005	Profit Centre	CEPC-DATAB	validFrom	YYYYMMDD to ISO	Valid date	ERR-MAP-002
MAP-PC-006	Profit Centre	CEPC-DATBI	validTo	YYYYMMDD to ISO	>= validFrom	ERR-MAP-002
MAP-ML-001	Material Ledger	MLDOC-MATNR	materialId	Preserve raw and display	Required	ERR-MAP-001
MAP-ML-002	Material Ledger	MARA-MAKTX	materialDescription	Trim	Optional	Business exception
MAP-ML-003	Material Ledger	MLDOC-WERKS	plant	Preserve	Required	ERR-MAP-001
MAP-ML-004	Material Ledger	MLDOC-BWART	movementType	Map to movement category	Known code	ERR-MAP-003
MAP-ML-005	Material Ledger	MLDOC-MENGE	quantity	Decimal	Numeric	ERR-MAP-006
MAP-ML-006	Material Ledger	MLDOC-MEINS	unitOfMeasure	ISO/UoM map	Known UoM	ERR-MAP-003
MAP-ML-007	Material Ledger	CKMLCR-PVPRS	periodicUnitPrice	Decimal	Numeric	ERR-MAP-006

ID	Domain	Source Field	Target Field	Transformation	Validation	Error Handling
MAP-ML-008	Material Ledger	MLDOC-BUDAT	postingDate	YYYYMMDD to ISO	Valid date	ERR-MAP-002
MAP-PO-001	Purchase Order	EKKO-EBELN	purchaseOrder	Preserve	Required	ERR-MAP-001
MAP-PO-002	Purchase Order	EKPO-EBELP	poLineItem	Left-pad	Required	ERR-MAP-001
MAP-PO-003	Purchase Order	EKKO-LIFNR	vendorId	Normalize	Required	ERR-MAP-001
MAP-PO-004	Purchase Order	EKKO-BEDAT	poDate	YYYYMMDD to ISO	Valid date	ERR-MAP-002
MAP-PO-005	Purchase Order	EKPO-MATNR	materialId	Preserve	Required for material PO	ERR-MAP-001
MAP-PO-006	Purchase Order	EKPO-MENGE	orderedQuantity	Decimal	>0	ERR-MAP-006
MAP-PO-007	Purchase Order	EKPO-NETPR	netPrice	Decimal	>=0	ERR-MAP-006
MAP-PO-008	Purchase Order	EKBE-BEWTP	poHistoryCategory	Map GR/IR/payment	Known code	ERR-MAP-003
MAP-SO-001	Sales Order	VBAP-VBELN	salesOrder	Preserve	Required	ERR-MAP-001
MAP-SO-002	Sales Order	VBAP-POSNR	salesLineItem	Left-pad	Required	ERR-MAP-001
MAP-SO-003	Sales Order	VBAP-KUNNR	customerId	Normalize	Required	ERR-MAP-001
MAP-SO-004	Sales Order	VBAP-AUDAT	orderDate	YYYYMMDD to ISO	Valid date	ERR-MAP-002
MAP-SO-005	Sales Order	VBAP-MATNR	materialId	Preserve	Required	ERR-MAP-001
MAP-SO-006	Sales Order	VBAP-KWMENG	orderedQuantity	Decimal	>0	ERR-MAP-006
MAP-SO-007	Sales Order	VBAP-NETWR	netValue	Decimal	>=0	ERR-MAP-006
MAP-SO-008	Sales Order	VBFA-VBTYP_N	documentFlowStatus	Map delivery/invoice/payment	Known code	ERR-MAP-003
MAP-FA-001	Fixed Assets	ANLA-ANLN1	assetNumber	Preserve	Required	ERR-MAP-001
MAP-FA-002	Fixed Assets	ANLA-ANLN2	assetSubnumber	Preserve	Required	ERR-MAP-001
MAP-FA-003	Fixed Assets	ANLA-TXT50	assetDescription	Trim	Required	ERR-MAP-001
MAP-FA-004	Fixed Assets	ANLA-AKTIV	capitalizationDate	YYYYMMDD to ISO	Valid date	ERR-MAP-002
MAP-FA-005	Fixed Assets	ANLC-KANSW	acquisitionValue	Decimal	Numeric	ERR-MAP-006
MAP-FA-006	Fixed Assets	ANLC-NAFAG	accumulatedDepreciation	Decimal	Numeric	ERR-MAP-006
MAP-BS-001	Bank Statement	FEBKO-KUKEY	statementKey	Preserve	Required	ERR-MAP-001
MAP-BS-002	Bank Statement	FEBKO-AZNUM	statementNumber	Preserve	Required	ERR-MAP-001
MAP-BS-003	Bank Statement	FEBEP-ESNUM	statementLine	Preserve	Required	ERR-MAP-001
MAP-BS-004	Bank Statement	FEBEP-WRBTR	amount	Decimal	Numeric	ERR-MAP-006
MAP-BS-005	Bank Statement	FEBEP-VALUT	valueDate	YYYYMMDD to ISO	Valid date	ERR-MAP-002
MAP-BS-006	Bank Statement	FEBEP-XBLNR	bankReference	Trim	Optional	None
MAP-BS-007	Bank Statement	FEBEP-AUGBL	clearingDocument	Null if unmatched	Optional	Business exception
MAP-BA-001	BudgetActual	ACDOCP-RBUKRS	companyCode	Preserve	Required	ERR-MAP-001
MAP-BA-002	BudgetActual	ACDOCP-RCNTR	costCentre	Preserve	Must exist	ERR-MAP-003
MAP-BA-003	BudgetActual	ACDOCP-RACCT	glAccount	Preserve	Must exist	ERR-MAP-003
MAP-BA-004	BudgetActual	ACDOCP-POPER	fiscalPeriod	Preserve	001-016	ERR-MAP-001
MAP-BA-005	BudgetActual	ACDOCP-HSL	budgetAmount	Decimal	Numeric	ERR-MAP-006
MAP-BA-006	BudgetActual	ACDOCA-HSL	actualAmount	Decimal	Numeric	ERR-MAP-006
MAP-BA-007	BudgetActual	Derived	varianceAmount	actual - budget	Numeric	ERR-MAP-006
MAP-BA-008	BudgetActual	Derived	variancePercent	variance / budget	Handle zero budget	ERR-MAP-006
MAP-INV-001	Inventory	MATDOC-MBLNR	materialDocument	Preserve	Required	ERR-MAP-001
MAP-INV-002	Inventory	MATDOC-ZEILE	materialDocLine	Preserve	Required	ERR-MAP-001
MAP-INV-003	Inventory	MATDOC-MATNR	materialId	Preserve	Required	ERR-MAP-001
MAP-INV-004	Inventory	MATDOC-WERKS	plant	Preserve	Required	ERR-MAP-001
MAP-INV-005	Inventory	MATDOC-LGORT	storageLocation	Preserve	Optional	None
MAP-INV-006	Inventory	MATDOC-BWART	movementType	Map receipt/issue/transfer	Known code	ERR-MAP-003
MAP-INV-007	Inventory	MATDOC-MENGE	movementQuantity	Decimal	Numeric	ERR-MAP-006
MAP-INV-008	Inventory	MATDOC-MEINS	unitOfMeasure	UoM lookup	Known UoM	ERR-MAP-003
MAP-INV-009	Inventory	MATDOC-DMBTR	movementValue	Decimal	Numeric	ERR-MAP-006
MAP-INV-010	Inventory	MATDOC-BUDAT	postingDate	YYYYMMDD to ISO	Valid date	ERR-MAP-002

Appendix B: Full Test Scenario Table

ID	Category/Title	Preconditions/Title	Steps/Validation	Expected Result
TST-FNC-001	Happy Path GL Extraction	100 GL entries in SAP sandbox	Trigger delta, verify Kafka count, load FinSight, reconcile	100 records loaded, checksum match
TST-FNC-002	AP Open Items Multi-Currency	INR, USD, EUR AP items and TCURR rates	Extract AP, convert currency, check ageing	Correct INR conversion and ageing
TST-FNC-003	Master Data Delta Sync	Existing cost centre sync	Create cost centre, run cycle, query target	New cost centre appears
TST-FNC-004	Multi-Company Code Batch	Entries for MC01, MC02, MC03	Extract mixed batch, verify routing	Each company routes to correct tenant
TST-FNC-005	Fiscal Period Mapping	SAP periods 001-016	Extract all periods, verify mapping	013-016 flagged as special periods
TST-FNC-006	7-Level Hierarchy Flatten	Cost centre hierarchy exists	Extract hierarchy and inspect target	Level1-Level7 populated
TST-FNC-007	P2P Flow Tracing	PO, GR, invoice, payment exist	Extract stages and trace flow	All stages linked
TST-FNC-008	Bank Statement Load	50 bank items	Load IDoc/CDS, transform, validate clearing	Matched cleared, unmatched open
TST-FNC-009	Budget vs Actual Variance	Budget and actuals for 5 centres	Extract both and calculate variance	Matches SAP CO report
TST-FNC-010	End-of-Day Full Reconciliation	Full day across domains	Run all extractions and recon	Zero unexplained breaks
TST-NFR-001	Performance	Peak 500K GL entries	Completes within 2-hour SLA	
TST-NFR-002	Concurrency	12 domains run together	No deadlocks or resource contention	
TST-NFR-003	Latency	500 concurrent FinSight requests	P95 under 5 seconds, controlled 429	
TST-NFR-004	Scalability	1K to 100K records	Linear throughput until documented ceiling	
TST-NFR-005	Endurance	24-hour operation	No memory leak or connection exhaustion	
TST-FLR-001	Failure	SAP RFC failure mid-batch	Circuit opens, partial batch preserved	
TST-FLR-002	Failure	FinSight HTTP 429	Honors Retry-After and succeeds eventually	
TST-FLR-003	Failure	Kafka broker failure	Failover with zero message loss	
TST-FLR-004	Failure	10 malformed records in 100	90 succeed, 10 DLQ	
TST-FLR-005	Failure	5-minute network partition	Buffers and recovers with eventual consistency	
TST-SEC-001	Security	OAuth token expires	Automatic refresh without data loss	
TST-SEC-002	Security	Invalid token	HTTP 401, audit log, no data exposure	
TST-SEC-003	Security	Encryption verification	TLS 1.2+ in transit, AES-256 at rest	
TST-REC-001	Reconciliation	INR 1.50 discrepancy	Break detected and alert generated	
TST-REC-002	Reconciliation	Missing cost centre	Referential check catches orphan	